Spatio-temporal analysis of TB infection rates amongst homeless shelter clients exposed to a highly infectious individual

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Background

In May 2008, an outbreak of TB began in a homeless shelter in British Columbia, Canada. The index case, diagnosed with AFB4+ pulmonary TB, used the shelter for 12 nights, seeding a large first wave of both active cases and latent TB infections (LTBI). By December 2013, the outbreak had grown to include 52 active cases, with 2,310 community members identified as potentially exposed and subsequently investigated with tuberculin skin testing (TST). **By combining a unique dataset of the shelter client roll calls and bed locations with a TST screening database, we were able to examine the influence of exposure duration and distance from the index case on other shelter clients' risk of TB infection.**

Methods

Bed map and roll call data : Each night, incoming shelter clients are assigned a numbered bunk; the bed number and client name are recorded on paper roll calls. As part of the outbreak investigation, the shelter provided the public health unit with the roll calls for the 12 nights the index case used the shelter, along with a scaled map of the numbered bed locations as the shelter was configured in early 2008. From these data, we identified individuals exposed to the index case at the shelter, their sleeping distance from the index case, and the number of nights spent in the shelter with the index case.

Client TB status definitions : We cross-referenced individuals exposed to the index case using a TST screening database and assigned shelter clients to one of three outcomes: "infected" (having a diagnosis of active TB or a positive TST result with no prior history of TB disease and no prior positive TST); "uninfected" (at least one negative TST two or more months after exposure to the index case); and "unknown" (clients who were not screened, or received a TST but did not have it read, or had a prior positive TST, or had prior TB infection, or had BCG vaccination that precluded definitively linking their positive TST to the index case).

Distance and Exposure Variables : For each client, we calculated the total **time exposed to the index case (*T*)** as the cumulative, non-consecutive, number of days that a client slept in the shelter when the index case was present. We also calculated each client's bed distance from the index case's bed (*D*) under three different scenarios: D_{sum} – the sum of the nightly distances; D_{avg}; – the average nightly distance; and D_{min} – the minimum. We also normalized D_{sum} by *T* to create a combined measure, D_T – the distance-to-exposure time ratio.

Statistical Methods: Odds ratios were calculated using a univariable logistic regression with the response variable infection status (Infected/Uninfected) and predictor variables (considered individually) *T*, D_{sum}, D_{avg}, D_{min}, and D_T. All predictors were treated as continuous, unless otherwise indicated. All tests for significance were two-sided and were considered significant if $p < 0.05$. Clients with unknown TB status were not included in the analysis. A threshold was established using recursive partitioning with a maximum tree depth of one using R v 3.0.2 and the rpart package v 4.1. The recursive partitioning took as input a continuous variable, such as exposure or proximity, and established a threshold that optimally separated infected and uninfected clients.

Results & Discussion

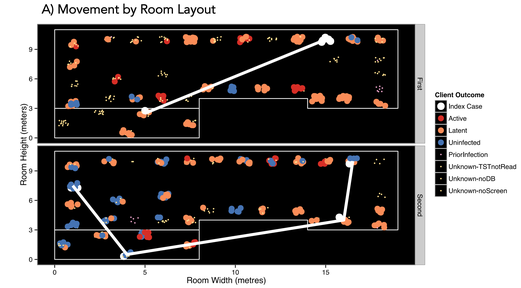
Cohort & visualizing client movement

Distribution of shelter clients' TB screening results. A total of 161 clients were exposed to the index case during their stay in the shelter. The index case had two visits: the first for seven nights over an 11-night window and the second four months later for five nights over a 10-night window. Of all clients, a total of 58 were screened, and 3 had previously known TB infections.

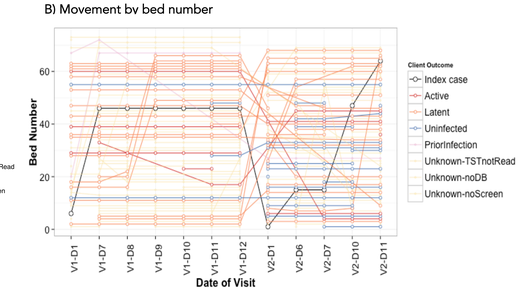
Outcome	Total	Both Visits	First Only	Second Only
Active	8	2	3	3
Latent	31	13	7	11
Not-Infected	19	3	1	15
Prior Infection	3	1	1	1
Total with Outcome	58	18	11	28
Total Unknowns	100	2	53	446
Total - All	161	21	64	76

Client movement over the index case's two shelter visits. Client movement was visualized two ways: **A) By room layout.** Dots indicate where a specific client slept each night and are coloured according to TB status. The index case is shown in white; the index case's movement is emphasized with a white line ; **B) By bed number.** Distance from the index case over time. The index case is shown in black. These results show that client sleeping location was more variable during the second index case visit to the shelter.

A) Movement by Room Layout



B) Movement by bed number

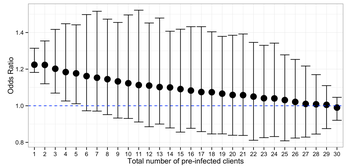


Assessing the impact of distance & exposure variables

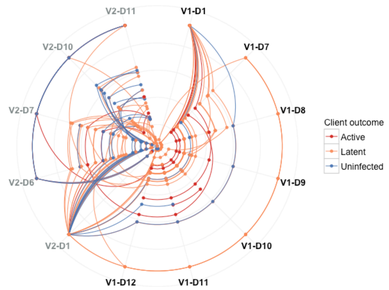
Odds ratios of TB infection (active and latent), according to time exposed to the index case and sleeping distance from the index case. *T*: cumulative nights exposed to the infectious index; D_{sum}: sum of the nightly sleeping distance from the index; D_{avg}: average nightly distance from the index; D_{min}: minimum distance from the index; D_T: distance-to-exposure time ratio. Only exposure time was found to be significant.

Variable	Odds Ratio (95% Confidence Interval)	p-value
T	**1.26 (1.05 - 1.58)**	**0.02**
D_{sum}	1.01 (1.00 - 1.03)	0.08
D_{avg}	0.95 (0.85 - 1.03)	0.33
D_{min}	0.95 (0.86 - 1.03)	0.21
D_T	0.95 (0.85 - 1.05)	0.33

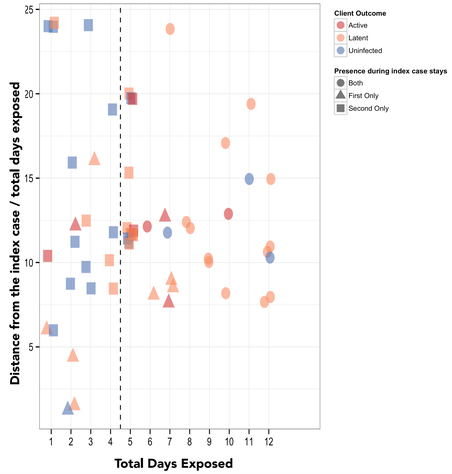
Sensitivity Analysis of Exposure Time Ratio. We varied the total number of cases that could have had a pre-existing latent infection to explore whether baseline LTBI rates affected our finding. Exposure time was found to be significant under up to ~5 pre-existing infections.



Distance from the index case. The closer a client is to the centre (bulls-eye), the closer to the index case. Individual client's visits are joined with a line to show the client's distance from the index case over time. A day is representing using V#-D#, where V1-D1, for example, is the first day (D1) of the index cases' first visit (V1). Non-crossing lines indicate consistent client sleeping location.



Visualizing the cumulative exposure threshold. Using data mining methods, we found that five days' exposure optimally separated infected (all active and latent) from non-infected clients. Clients are plotted according to their cumulative exposure time T and their distance-to-exposure time ratio D_T. The dotted line indicates the maximally separating exposure time threshold determined via machine learning that best separates infected and uninfected clients.



Total time exposed to the index case was an important determinate of TB status, but sleeping proximity to the index case was not

Clients that were exposed to the index case for 5 days or greater had a higher odds of being infected with TB

The small sample size and incomplete medical history of our study necessitates further follow-up to verify our conclusions

Our approach may be helpful to other shelter environments with similar data constraints

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